

Step 1: Forward Diffusion (Graph Fragmentation)

Simulating Topological Destruction

Discrete Edge Corruption

Instead of continuous coordinate noise, NeuroDiffusion uses a **discrete diffusion process** over the graph topology $G = (V, E)$.

- **Initial State ($t=0$):** The neuronal tree is fully connected.
 - **Corruption Process:** Edges are progressively dropped via a Bernoulli sampling process controlled by a noise schedule $\bar{\alpha}_t$.
 - **Final State ($t \rightarrow T$):** The fully connected tree fractures into isolated, disconnected sub-trees (mimicking real-world tracing artifacts).
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Mathematical Formulation

The transition probability for an edge existing at timestep t is given by:

$$q(E_t | E_0) = \text{Bernoulli}(E_t; \bar{\alpha}_t \cdot E_0)$$

Where: - E_0 is the pristine ground-truth edge set. - $\bar{\alpha}_t$ decays from 1.0 to $(1.0 - \beta)$ according to a predefined schedule (e.g., linear or cosine).}

Examples of Fragmented Topologies

Corrupted Trees

Hyperparameter Tuning: β_{end}

Parameter: `forward_diffusion.beta_end` - This controls the terminal probability of an edge being dropped. - **High β_{end} :** Results in severely fragmented trees with massive gaps. Exposes the model to very hard examples during training, forcing it to learn long-range topological rules. - **Low β_{end} :** Useful for fine-tuning on mostly-intact datasets where fragments are closely clustered.